

Module: Corporate Finance & Equity Valuation

School of Economics

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**Valuation of Unlisted companies Using Relative
Valuation**

Learning Objectives

- Understand the different approaches to valuation
- Understand the basis of all valuation approaches
- Understand the essence of relative valuation
- Be able to value ordinary shares using the relative valuation **approaches** or price multiples.
- Understand the adjustments necessary to earnings and the comparable price multiple when using relative valuation method.

Approaches to Valuation

1. **Intrinsic valuation**, relates the value of an asset to its capacity to generate cash flows and the risk in these cash flows. In its most common form, intrinsic value is computed with a discounted cash flow valuation, with the value of an asset being the present value of expected future cash flows on that asset.

2. **Relative valuation or Pricing**, estimates how much to pay for an asset by looking at what others are paying for comparable assets, scaled to a common metric that they all share (earnings, revenues, subscribers).

3. **Contingent claim (or real option) valuation**, augments the value of assets, whose cash flows are contingent on an event happening.

Basis for all valuation approaches

- ❑ The use of valuation models in investment decisions are based on:
 - a perception that markets are inefficient and make mistakes in assessing value
 - an assumption about how and when these inefficiencies will get corrected
- ❑ In an efficient market, the market price is the best estimate of value. The purpose of any valuation model is then the justification of this value.
- ❑ The differences between different investment philosophies, all claiming to beat the market, lie in what types of inefficiencies they see in markets and how those inefficiencies get corrected.

The Essence of Relative Valuation (Pricing)

- ❑ In relative valuation, the value of an asset is compared to the values assessed by the market for similar or comparable assets.
- ❑ To do relative valuation, (what do we need?)
 - we need to identify comparable assets and obtain market values for these assets
 - convert these market values into standardized values, since the absolute prices cannot be compared. This process of standardizing creates price multiples.
 - compare the standardized value or multiple for the asset being analyzed to the standardized values for comparable asset, controlling for any differences between the firms that might affect the multiple, to judge whether the asset is under or over valued

Advantages of Relative Valuation (own study)

- ❑ **In sync with the market**: Relative valuation is much more likely to reflect market perceptions and moods than discounted cash flow valuation. This can be an advantage when it is important that the price reflect these perceptions as is the case when:
 - the objective is to sell an asset at that price today (IPO, M&A)
 - investing on “momentum” based strategies
- ❑ **With relative valuation**, there will always be a significant proportion of securities that are under valued and over valued. Since portfolio managers are judged based upon how they perform on a relative basis (to the market and other money managers), relative valuation is more tailored to their needs
- ❑ **Relative valuation** generally requires less explicit information than discounted cash flow valuation.
- ❑ **In relative valuation**, you are playing the “incremental” game, where you hope to make money by getting the next increment (earnings report, news story etc.) right.

Disadvantages of Relative Valuation (own study)

- ❑ A portfolio that is composed of stocks which are under valued on a relative basis may still be overvalued, even if the analysts' judgments are right. It is just less overvalued than other securities in the market.
- ❑ Relative valuation is built on the assumption that markets are correct in the aggregate, but they make mistakes on individual securities. To the degree that markets can be over or under valued in the aggregate, relative valuation will fail.
- ❑ Relative valuation may require less information in the way in which most analysts and portfolio managers use it. However, this is because implicit assumptions are made about other variables (that would have been required in a discounted cash flow valuation). To the extent that these implicit assumptions are wrong the relative valuation will also be wrong.

When relative valuation works best.

□ This approach is easiest to use when:

- there are a large number of assets comparable to the one being valued
- these assets are priced in a market
- there exists some common variable that can be used to standardize the price

□ This approach tends to work best for investors:

- who have relatively short time horizons
- Who are judged based on a relative benchmark (the market, other portfolio managers following the same investment style etc.)
- Who can take actions that can take advantage of the relative mispricing. For instance, a hedge fund can buy the under valued and sell the over valued assets. Note that: a long-only investor cannot (i.e. an investor who buys only).

Relative valuation is pervasive...

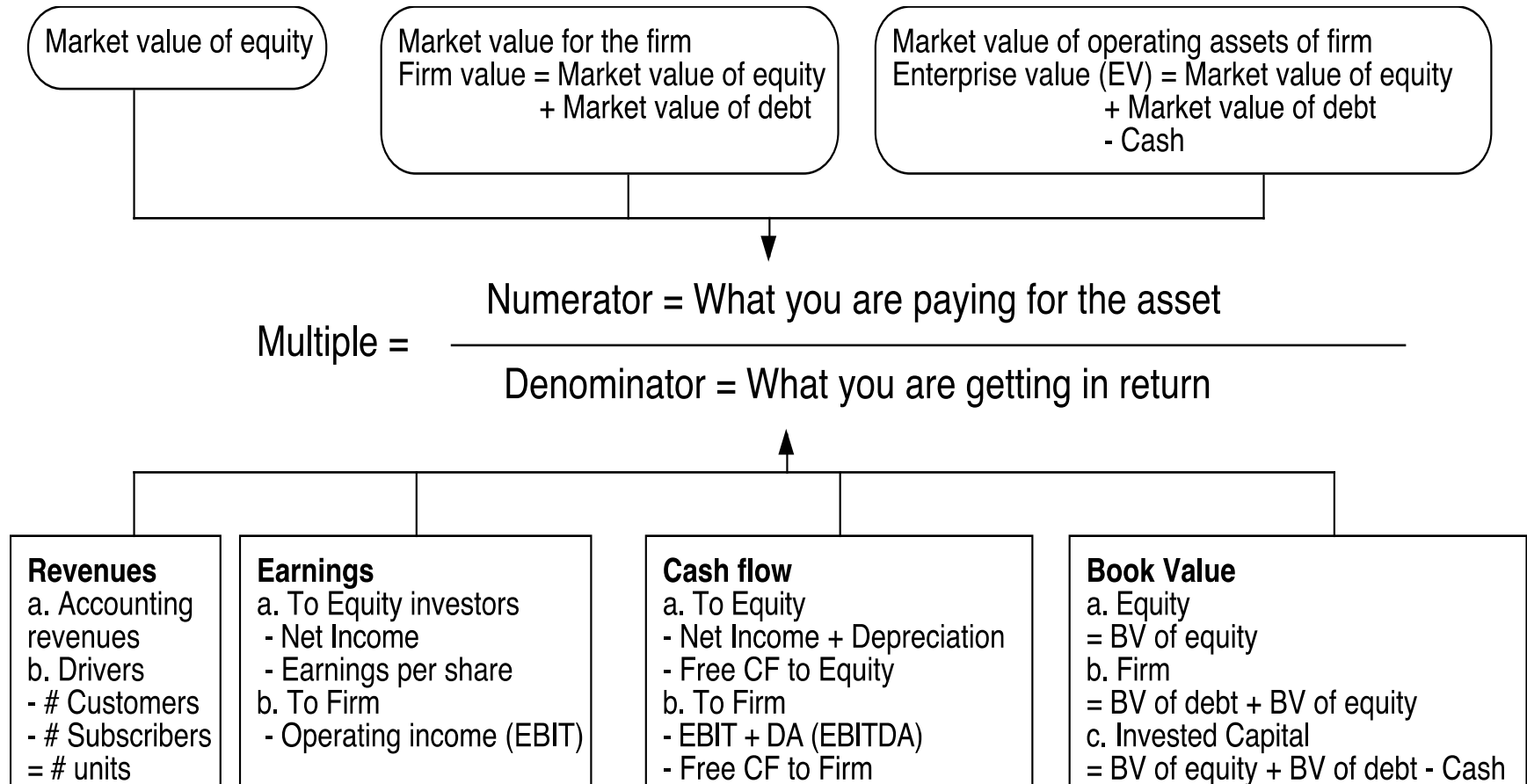
- ❑ Relative valuation is pervasive in that this method of valuing companies is extremely common and widespread throughout the financial industry.
- ❑ Most asset valuations are relative.
- ❑ Key implications of its pervasiveness are:
- ❑ Dominant approaches on most equity valuations on Wall Street.
 - Almost 85% of equity research reports use comparables and a multiple such as P/E ratio, EV/EBITDA to justify their recommendations.
 - More than 50% of all mergers and acquisitions valuations are based upon multiples
 - Rules of thumb based on multiples are not only common but are often the basis for final valuation judgments.
- ❑ While there are more discounted cashflow valuations in consulting and corporate finance, they are often relative valuations masquerading as discounted cash flow valuations.
 - The objective in many discounted cashflow valuations is to back into a number that has been obtained by using a multiple.
 - The terminal value which is a significant number of discounted cashflow valuations is estimated using a multiple. (Final year metric x exit multiple)

The Market Imperative....

Since relative valuation models more closely mirror the market:

- ❑ Relative valuation is much **more likely to reflect market perceptions and moods than discounted cash flow valuation**. This can be an advantage when it is important **that the price reflects these perceptions as is the case when:**
 - the objective is **to sell a security at that price today** (as in the case of an IPO)
 - investing on **“momentum” based strategies**
- ❑ With relative valuation, there will always be a significant proportion of securities that are under valued and over valued.
- ❑ Since **portfolio managers are judged based upon how they perform on a relative basis** (to the market and other money managers), **relative valuation is more tailored to their needs**.
- ❑ Relative valuation generally **requires less information than discounted cash flow valuation** (especially when multiples are used as screens)

Multiples are just standardized estimates of price...



The Four Steps to Deconstructing Multiples

❑ Define the multiple

- In use, the same multiple can be defined in different ways by different users. When comparing and using multiples, estimated by someone else, it is critical that we understand how the multiples have been estimated

❑ Describe the multiple

- Too many people who use a multiple have no idea what its cross sectional distribution is. If you do not know what the cross sectional distribution of a multiple is, it is difficult to look at a number and pass judgment on whether it is too high or low.

❑ Analyze the multiple

- It is critical that we understand the fundamentals that drive each multiple, and the nature of the relationship between the multiple and each variable.

❑ Apply the multiple

- Defining the comparable universe and controlling for differences is far more difficult in practice than it is in theory.

Definitional Tests

❑ Is the multiple consistently defined?

- Proposition 1: Both the value (the numerator) and the standardizing variable (the denominator) should be to the same claimholders in the firm. In other words, the value of equity should be divided by equity earnings or equity book value, and firm value should be divided by firm earnings or book value.

❑ Is the multiple uniformly estimated?

- The variables used in defining the multiple should be estimated uniformly across assets in the “comparable firm” list.
- If earnings-based multiples are used, the accounting rules to measure earnings should be applied consistently across assets. The same rule applies with book-value based multiples.

Example 1: Price Earnings Ratio: Definition

$$PE = \text{Market Price per Share} / \text{Earnings per Share}$$

- ❑ There are a number of variants on the basic PE ratio in use. They are based on how the price and the earnings are defined.

Price: is usually the current market price

is sometimes the average price for the year

EPS: EPS in most recent financial year

EPS in trailing 12 months

Forecasted earnings per share next year

Forecasted earnings per share in future year

- ❑ This is only the tip of the iceberg. You can have EPS before and after extraordinary items, primary and diluted EPS..
- ❑ When you are negotiating with someone else and you are both using PE ratios to make your case, the first step is to make sure that you are using the same PE ratio.
- ❑ There is also the tendency on the part of analysts to pick the definition of PE that best fits their biases. For instance, bullish analysts in the 1990s almost always used forward PE whereas bearish analysts used trailing PE. Since earnings were rising the former were generally much lower than the latter.

Example 2: Still on PE ratios

- ☐ Assuming that you are comparing the PE ratios across technology companies, many of which have options outstanding. What measure of PE ratio would yield the most consistent comparisons?
 - a. Price/ Primary EPS (actual shares, no options)
 - b. Price/ Fully Diluted EPS (actual shares + all options)
 - c. Price/Partially Diluted EPS(counting only in-the-money options)
 - d. Other

- ☐ The answer is actually none of the above. You do have to control for the number of options outstanding... So using primary EPS is out of the question. The other three approaches treat all options as equivalent (even the one that counts in-the-money options only treats them all the same). To get consistency, you would have to value the options as options:
- ☐
$$\text{PE ratio} = (\text{Market Capitalization} + \text{Estimated Value of Options Outstanding}) / \text{Net Income}$$

Example 3: Enterprise Value /EBITDA Multiple

- ❑ The enterprise value to EBITDA multiple is obtained by netting cash out against debt to arrive at enterprise value and dividing by EBITDA.
- ❑ Note that cash reduces the cost of acquiring the company.

$$\frac{\text{Enterprise Value}}{\text{EBITDA}} = \frac{\text{Market Value of Equity} + \text{Market Value of Debt} - \text{Cash}}{\text{Earnings before Interest, Taxes and Depreciation}}$$

1. Why do we net out cash from firm value? (see note above)
 2. What happens if a firm has cross holdings which are categorized as:
 - Minority interests?
 - Majority active interests?
- ❑ The problem with using firm value is that **cash is included in the numerator but not in the denominator**. That is why the enterprise value version makes more sense...
 - ❑ A broad problem is posed when firms have holdings in other firms. **If such holdings are passive**, Value to EBITDA **multiples will be overstated**, since the numerator will include the value of your holdings, while the EBITDA will not include any of the income from these holdings. If such holdings **are majority active** and consolidated, the value to EBITDA **will be understated** because the numerator will include only the portion of the equity you own in the subsidiary but the EBITDA will include all of the EBITDA in the subsidiary.
 - ❑ **The safest thing to do** (assuming you can do this) is to net out the market value of your holdings from the numerator (for both active and passive holdings) and the EBITDA of your holdings from the denominator (for majority active holdings)

Example 4: A Housing Price Multiple

The bubbles and busts in housing prices has led investors to search for a multiple that they can use to determine when housing prices are getting out of line. One measure that has acquired adherents is the ratio of housing price to annual net rental income (for renting out the same house). Assume that you decide to compute this ratio and compare it to the multiple at which stocks are trading. Which valuation ratio would be the one that corresponds to the house price/rent ratio?

- a. Price Earnings Ratio
- b. EV to Sales
- c. EV to EBITDA
- d. EV to EBIT

☐ Net Rental income is closest to EBIT, as long as you don't net out interest expenses and the house price is closest to EV. So. **EV/EBIT is your best match.**

Valuation of Ordinary Share: P/E Ratio

- P/E ratio = Market price per share/EPS
- **The valuation of ordinary shares using the P/E ratio requires three basic steps:**
 - First calculate future sustainable EPS of the company being valued.
 - **Second, choose a comparable listed company and adjust its P/E ratio for different risk and growth factors or use an adjusted sector average P/E ratio.**
 - Finally, calculate the value per ordinary share using the P/E ratio.
- **The following formula is used:**

Value per ord. share = Comparable P/E Ratio x Sustainable future EPS

Valuation of Ordinary Share : P/E Ratio

Example 5

Padua (Pty) Ltd is not a listed company. Its future sustainable EPS is 25c. A comparable listed company has EPS of 60c and price per share of 600c. Calculate the value of each ordinary share of Padua (Pty) Ltd.

Example 5. Answer

- The comparable company's P/E ratio is:
 - **$(600/60) = \underline{10}$ (Meaning?)**
- The value per ordinary share is:
 - **$25c \times 10 = \underline{250c}$**

Sustainable Future EPS of the company being Valued:

- Note that sustainable (or maintainable) future EPS are used.
- The sustainable future EPS are adjusted for the following:
 - Exceptional, abnormal, unusual, or non-recurring items.
 - Profit or loss on discontinued operations.
 - Adjustment of expenses to realistic levels (e.g. management fees).
 - Adjustment of “non-operating” income or expenses.

Valuation of Ordinary Share: P/E Ratio

COMPARABLE P/E RATIO:

- The P/E ratio of a comparable company (or sector average P/E ratio) must be adjusted for differences in risk and growth factors.
- If the target company is MORE risky than the comparable listed company, then the P/E ratio needs to be REDUCED, and vice-versa.
- If the target company has a higher growth, the P/E ratio would be increased.
- **The risk factors adjusted for include the following:**
 - Marketability (unlisted company more risky).
 - Liquidity (unlisted company more risky).
 - Access to finance (more difficulty for an unlisted company).
 - Management styles.
 - Reliance on a small number of key employees
 - The relative size of the entities.
 - Size of the shareholding (minority vs controlling interests).
 - Leverage and quality of earnings.
 - The nature (e.g., highly specialized – more risky) and relative age of NCAs.
 - The diversity of the product ranges.

Ordinary Share Valuation: P/E Ratio

- Example 6. Padua Pty Ltd has 100 000 shares in issue. The actual financial results for 31 May 2020 are shown below:

Income statement for the year ended 31 May 2020		Rand
Revenue		1 000 000
Less. Cost of sales		<u>(500 000)</u>
Gross Profit		500 000
Operating expenses		(110 000)
Finance costs		<u>(30 000)</u>
Profit before taxation		360 000
Taxation		<u>(104 400)</u>
Profit for the year from continuing operations		255 600
Loss for the year from discontinued operations		<u>(10 000)</u>
Profit for the year		245 600

The management team of the most comparable listed firm is considered to be “cowboys” while Padua Pty Ltd’s management team is considered conservative. The profit from continuing operations is considered sustainable in the future.

Required:

1. Value the shares of Padua Pty Ltd using the P/E valuation method. Assume that the P/E ratio of the comparable company is 10.
2. What factors must be considered when interpreting the results?

1. # shares = 100k

Value = price

P/E of comparable = R10

Adjustments	conservative	+1
	unlisted	-2
	lack of liquidity	-2

Adjusted P/E = 7

$$P / \frac{265\ 600}{245\ 600} = 7$$

$$P = 1\ 719\ 200 \div \text{\# shares}$$
$$= R17,19 \text{ per share } \approx R17,89$$

2. • If P/E is an appropriate comparison

- Are the adjustments for risk appropriate?
- If the compared company sufficiently comparable?
- The method is very simplistic. Use another as sanity check
- Are current earnings indication of future CFs?
- Do a/c profits represent future CFs?
- What about differences in growth prospects?

Ordinary Share Valuation: P/E Ratio

SUGGESTED SOLUTION TO EXAMPLE 6

PART 1

- Sustainable future earnings: R255 600.
- Adjusting P/E ratio of comparable listed company:

	P/E
Comparable listed company	10
Adjustments for differences in risk:	
Not listed: Difficult to raise capital	-2
Not listed: Lack of liquidity/marketability	-2
Management conservative	+1
Adjustment for differences in growth:	-
ADJUSTED P/E ratio to be used	<u>7</u>

Value of Padua shares = $R255\,600 \times 7 =$ **R1 789 200**

Value per share = $R1\,789\,200 / 100\,000 =$ **R17.89**

Ordinary Share Valuation: P/E Ratio

SUGGESTED SOLUTION TO EXAMPLE 6

PART 2

The following factors would be taken into account:

- Are the adjustments for risk appropriate?
- Is the comparable listed company sufficiently comparable?
- The very simplistic method. Use another method as sanity check.
- Are current earnings an indication of future cash flows?
- Do accounting profits represent future cash flows?
- What about differences in growth prospects?

Other Comparable Valuation Methods

MARKET-TO-BOOK RATIO:

- Book value (NAV) per share (for the target firm) = Total equity (excluding None Controlling Interest)/No. of ord. shares.
- Market-to-book (for the comparable firm) = Price per share/Book value per share.
- **Value of ordinary shares = Book value per share (of target firm) x comparable market-to-book ratio.**
- High growth firms have high market-to-book ratios.

PRICE-TO-SALES RATIO:

- Price-to-sales ratio = Price per share/Sales per share
- **Value of ordinary shares = Sales per share (of target firm) x comparable price-to-sales ratio.**
- Useful when a company is making losses but is growing sales.

PRICE-TO-CASH EARNINGS RATIO:

- Price-to-cash ratio = Price per share/Cash earnings per share.
- Cash earnings = Profit after tax + depreciation.
- **Value of ordinary shares = Cash earnings per share (target firm) x comparable price-to-cash earnings ratio.**

Enterprise Value: The EV/EBITDA ratio

Determining Enterprise Value Using the P/E Framework.

- Because the P/E ratio is sensitive to different capital structures (or leverage) many analysts use the enterprise value/EBITDA ratio of a comparable listed company to estimate the total value of the firm being valued.
- The firm (or enterprise) value of a comparable listed company is determined as follows:
 - $V_{\text{Firm}} = V_{\text{Debt}} + V_{\text{Equity}}$
- The total value of a firm being valued is determined as follows:
 - $V_{\text{Firm}} = (\text{Ent.Value/EBITDA})_{\text{Comparable}} \times \text{EBITDA}_{\text{Firm being valued}}$

EXAMPLE 7

- Gisimu (Pty) Ltd's debt is valued at R4.25m and its EBITDA is R3.50m. Gitsalina Ltd is a listed company that is comparable to Gisimu Pty Ltd. Gitsalina Ltd's share price is R10.60, its debt is valued at R11.62m, its EBITDA is R10.80m and it has 9.5m shares in issue. Required: Using the **enterprise value/EBITDA** multiple, determine the enterprise value of Gisimu (Pty) Ltd.

Enterprise Value: The EV/EBITDA ratio

- Example 7
- Gisimu (Pty) Ltd's debt is valued at R4.25m and its EBITDA is R3.50m while its shares in issue are 6,43million. Gitsalina Ltd is a listed company that is comparable to Gisimu pty ltd. Gitsalina Ltd's share price is R10.60, its debt is valued at R11.62m, its EBITDA is R10.80m and it has 9.5m shares in issue. Required: Using the **enterprise value/EBITDA** multiple, determine the enterprise value of Gisimu (Pty) Ltd.

Gisimu (unlisted)
 $V_d = 4,25m$
 EBITDA = 3,50m
 # shares = 6,43m

Gitraling (listed)
 value = R10,6
 $V_d = 11,62$
 EBITDA = 10,8m
 # shares = 9,5m

$$\frac{\text{value}}{\text{EBITDA}} = \frac{R10,6 \times 9,5 + 11,62}{10,8m}$$

Gitraling: 9,32
 Adjustments:
 unlisted - 2
 limited liquidity - 2

 5,32

$$= 9,32 \times 10,4$$

$$\frac{\text{Value}}{3,5m} = \frac{10,4}{5,32}$$

$$\text{Value} = R18,62m \times 36,4$$

$$\begin{aligned} &\text{- Debt } (4,25) \\ &\text{Equity } \times \\ &\quad \div \text{ # shares} \\ &= R5 \text{ per share} \end{aligned}$$

Enterprise Value: The EV/EBITDA ratio

ANSWER TO EXAMPLE 7

- Enterprise value of Gitsalina Ltd:

- $V_{\text{Gitsalina}} = V_{\text{Debt}} + V_{\text{equity}}$

- $V_{\text{Gitsalina}} = \text{R11.62m} + (\text{R10.60} \times 9.5\text{m}) = \underline{\text{R112.32m}}$

$$\text{EV/EBITDA}_{\text{Gitsalina}} = \text{R112,32m} / \text{R10,8m} = \text{R10,4}$$

- The total value of Gisimu (Pty) Ltd:

- $V_{\text{Gisimu}} = (\text{Enterprise Value/EBITDA})_{\text{Gitsalina}} \times \text{EBITDA}_{\text{Gisimu}}$

- $V_{\text{Gisimu}} = (\text{R112.32m} / \text{R10.8m}) \times \text{R3.50m} = \underline{\text{R36.40m}}$

- OR

- $\underline{V_{\text{Gisimu}}} = \text{R10,4} \times \text{R3,5m} = \underline{\text{R36,4m}}$

ANSWER TO EXAMPLE 7

- Given that Gisimu (pty) has 6,43 shares in issue, the value of its ordinary share should therefore be:

Value of Gisimu (pty)	R36,4m
Less: debt	<u>(R4,25m)</u>
Value of Equity _{Gisimu}	R32,15m
No. of shares in issue	<u>6,43m</u>
Value of the Gisimu share	R5,00

Example 8

- Let's walk through an example together of how to calculate a company's EBITDA multiple. ABC Wholesale Corp has a Market Cap of R69.3B as of March 1, 2018, a cash balance of R0.3B, and debt of R1.4B as of December 31, 2017. For the full year of 2017, its EBITDA was reported at R5.04B and the current analyst consensus estimate for 2018 EBITDA is R5.5B. What are the resulting historical and forward-looking multiples?
- Here are the steps to answer the question:
 1. Calculate the Enterprise Value (Market Cap plus Debt minus Cash) = $R69.3 + R1.4 - R0.3 = R70.4B$
 2. Divide the EV by 2017 EBITDA = $R70.4 / R5.04 = 14.0x$
 3. Divide the EV by 2018 EBITDA = $R70.4 / R5.50 = 12.8x$